

Sparse nonattacking chess placements

Every fixed asymptotic order and growth-index inversion

2 October 2026

Abstract

The numbers of ways to place n nonattacking kings or knights on an $n \times n$ board have the same classical leading equivalent, $e^{-9/2}n^{2n}/n!$. Their first relative corrections are different: $7/(3n)$ for kings and $37/(3n)$ for knights. We compute four corrections exactly and give a finite algorithm and a uniform remainder proof for every fixed order. The method applies to every fixed finite symmetric move set, including a uniform extension with k/n in a compact positive interval. Exact boundary contributions and triangle density explain the first correction. We also derive a Lambert- W growth-index inverse and state the necessary integer-rounding qualification. The leading law and general cluster-expansion methods are prior work; the results presented here are explicit specialized expansions with controlled errors.

1 Definitions and main results

Let K_n count subsets of n squares of the ordinary $n \times n$ board containing no attacking pair of kings, and let N_n be the corresponding knight count. These are OEIS [A201513](#) and [A201540](#). Rotations and reflections remain distinct. There is no one-piece-per-row or one-piece-per-column constraint.

Kotěšovec's leading equivalent, dated 29 November 2011 and reproduced in his book [2], is

$$K_n \sim N_n \sim B_n, \quad B_n = e^{-9/2} \frac{n^{2n}}{n!}. \quad (1)$$

It is not a new result of this report. The book also contains fixed-piece polynomial formulas. Pages 77 and 293 display the diagonal leading laws; pages 685–686 give the general bounded-move leading argument. A fixed-piece polynomial expansion alone does not justify substituting a growing piece count in its remainder. We supply that uniform step below.

Theorem 1. *Both sequences have rational relative expansions to every fixed order, with the corresponding next-power remainder. In particular,*

$$\frac{K_n}{B_n} = 1 + \frac{7}{3n} + \frac{119}{9n^2} + \frac{11327}{810n^3} + \frac{90475}{1944n^4} + O(n^{-5}), \quad (2)$$

$$\frac{N_n}{B_n} = 1 + \frac{37}{3n} + \frac{914}{9n^2} + \frac{330917}{810n^3} + \frac{966583}{1944n^4} + O(n^{-5}). \quad (3)$$

Consequently,

$$\frac{N_n}{K_n} = 1 + \frac{10}{n} + \frac{65}{n^2} + \frac{332}{3n^3} - \frac{4841}{6n^4} + O(n^{-5}). \quad (4)$$

The logarithmic corrections are particularly simple:

$$\log \frac{K_n}{B_n} = \frac{7}{3n} + \frac{21}{2n^2} - \frac{379}{30n^3} - \frac{357}{40n^4} + O(n^{-5}), \quad (5)$$

$$\log \frac{N_n}{B_n} = \frac{37}{3n} + \frac{51}{2n^2} - \frac{6559}{30n^3} - \frac{1397}{40n^4} + O(n^{-5}). \quad (6)$$

If the carrier is instead $e^{-9/2}(en)^n/\sqrt{2\pi n}$, the first relative corrections are $9/(4n)$ and $49/(4n)$. The carrier must therefore be specified when quoting a coefficient.

The mechanism is an application of established cluster-expansion ideas. Canonical independent-set cluster expansions and convergent tails are developed, for example, by Davies, Jenssen, and Perkins [3], who credit earlier canonical work of Pulvirenti and Tsagkarogiannis [4]. We do not claim a new general cluster method, or exhaustive historical priority for each specialized coefficient. The proof here is self-contained and the coefficient computation is exact.

[write] Added 2 October 2026, batch 77. This report is a single manuscript, printed in full: manuscript 03 of batch 77 of ProveIt’s incoming-reports intake, delivered in commit 096ee7b87 as an archive whose name, wrapper directory and main TeX file all have the stem **Sparse_Chess_Asymptotics** (the main file is now **article.tex**), and placed in commit d0e6008d9 among the OEIS-sequence asymptotics reports of the research-report collection. Its author line is empty and the delivery names no tool. It names no ProveIt commit, so it has no pin; its closing literature-scope paragraph mentions “targeted checks of the current relevant ProveIt catalog”, an unpinned, bounded search. The intake’s own untruncated search of the collection, the transseries volumes, **0eis/** and **docs/reports/** is the authoritative one: it found none of A201513, A201540, A201511, A201861 and no report on non-attacking placements, so the manuscript’s “did not reveal overlap” stands. Nothing here is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement. The writing step prefixed every label with **scp:**, added section labels and these **[write]** notes and two references [5, 6]; no statement, proof, number or table of the manuscript was changed.

Three carriers. Every coefficient of this report is relative to a named carrier, and the three in use give different numbers. (i) The coefficients c_r of Theorems 1 and 2 and of (2)–(6) are relative to $B_n = e^{-9/2}n^{2n}/n!$ of (1), with the factorial kept exact: first corrections $7/3$ (kings) and $37/3$ (knights). (ii) Relative to the Stirling-expanded carrier $e^{-9/2}(en)^n/\sqrt{2\pi n}$ the first corrections are $9/4$ and $49/4$, as stated above. (iii) The logarithmic model (27) of Section 7 expands $\log n!$ as well, so its $d_1 = 9/4$, $49/4$ are logarithmic coefficients relative to carrier (ii), equal to $c_1 - \frac{1}{12}$. Quoting a c_r against carrier (ii), or a d_j against B_n , is the tempting false reading.

Not the rook paths of a neighbouring report. The batch-77 report **a227578-ordered-rook-paths** counts lattice paths made of rook *moves*; the collection’s rook polynomials (**a330266-balanced-smirnov-poisson**) count non-attacking rook placements. Neither concerns kings or knights, and no result is shared.

Notation. Several letters carry two meanings. Each symbol is printed as delivered; the table fixes its meaning.

symbol	meaning here (and what it is <i>not</i>)
B_n, B	B_n is the carrier (1); $B = \frac{1}{2} \sum_{u \in S} (u_1 + u_2)$ is the boundary loss of Section 2.
$K_n, N_n, N(y)$	K_n (kings) and N_n (knights) are the counts; $N(y)$ is the integer threshold of Section 7, for either sequence.
d, d_j	$d = S $ is the number of moves ($d = 8$ for both pieces); $d_1, d_2, \dots$ are the logarithmic coefficients of (27), relative to carrier (ii) above.
$a, a(n, k)$	$a = \alpha_2 = -(d+1)/2$ ($= -9/2$ for kings and knights); $a(n, k)$ counts independent k -sets.
D	the operator $t d/dt$ of Section 5; the slope $D = \log x + 2$ of Section 7.
w, h	$w(T), h(T)$ are the width and height of a support; $w_j(T)$ is its Möbius weight (14); h_ℓ are the polynomials (21).
J	a compact θ -interval in Theorem 2; a cluster truncation order in Section 4.
A, A_n	$A = m(m-1)/2$ in the proof of Proposition 1; A_n is either main sequence in Section 7.
H, H_r	a finite graph H ($L_j(H), I_H$); the polynomials $H_r(t; \theta)$ and the function $H_{n,k}(t)$ of Section 5.
L, L_j	$L = \log y$ (Section 7); $L_j(H)$ are logarithm coefficients (Section 3).
x, y	lattice coordinates in Section 3; in Section 7, y is a target value and $x = L/W(eL)$ the Lambert core.
W	the positive real (principal) Lambert branch W_0 .
$c_r, c_r(\theta)$	$c_r = c_r(1)$; $c_1^K(\theta), c_1^N(\theta)$ of Section 6 are (13) specialized to kings and knights.

2 The finite-range theorem and its geometry

Fix a nonempty finite symmetric set $S \subset \mathbb{Z}^2 \setminus \{0\}$, and write $d = |S|$. Let G_n have vertex set $V_n = \{0, \dots, n-1\}^2$, with x, y adjacent when $y - x \in S$. Write

$$I_n(z) = \sum_{k \geq 0} a(n, k) z^k, \quad \log I_n(z) = \sum_{j \geq 1} b_j(n) z^j. \quad (7)$$

Here $a(n, k)$ counts independent k -subsets. We will prove that for each fixed j there are rational numbers $\alpha_j, \beta_j, \gamma_j$ for which

$$b_j(n) = \alpha_j n^2 + \beta_j n + \gamma_j \quad (8)$$

for all sufficiently large integers n . In particular

$$(\alpha_1, \beta_1, \gamma_1) = (1, 0, 0), \quad \alpha_2 = -\frac{d+1}{2}.$$

Put $a = \alpha_2$.

Theorem 2. *For every fixed $M \geq 0$,*

$$a(n, n) = \frac{n^{2n}}{n!} e^a \left(\sum_{r=0}^M \frac{c_r}{n^r} + O_{M,S}(n^{-M-1}) \right), \quad (9)$$

where $c_0 = 1$ and every c_r is rational. A finite formula for c_r uses only the cluster polynomials $b_2, \dots, b_{r+2}$.

More generally, for any fixed compact interval $J \subset (0, \infty)$, uniformly over integers k with $\theta = k/n \in J$,

$$a(n, k) = \frac{n^{2k}}{k!} e^{a\theta^2} \left(\sum_{r=0}^M \frac{c_r(\theta)}{n^r} + O_{M,S,J}(n^{-M-1}) \right). \quad (10)$$

The coefficient functions have the explicit finite formula in Section 5.

The actual value $\theta = k/n$ is used in (10). Thus a choice $k = \lfloor \lambda n \rfloor$ retains its fractional-part corrections. No uniformity as $\theta \rightarrow 0$ or $\theta \rightarrow \infty$ is asserted.

2.1 Boundary loss and triangles

Define

$$B = \frac{1}{2} \sum_{u \in S} (|u_1| + |u_2|), \quad \tau = \frac{1}{6} \# \{(u, v) \in S^2 : v - u \in S\}.$$

Thus τ is the number of triangles per bulk vertex, when each triangle is counted once. The first correction is

$$\boxed{c_1(1) = B + \frac{1}{3} - \tau.} \quad (11)$$

For kings $(d, B, \tau) = (8, 6, 4)$; for knights $(d, B, \tau) = (8, 12, 0)$. The degree determines the common leading term, but the boundary loss and triangles distinguish the next term.

To verify (11), the elementary graph identities are

$$\begin{aligned} b_2 &= -\frac{|V|}{2} - |E|, \\ b_3 &= \frac{|V|}{3} + 2|E| + \sum_{v \in V} \binom{\deg(v)}{2} - \#\{\text{triangles}\}. \end{aligned}$$

Since

$$|E(G_n)| = \frac{1}{2} \sum_{u \in S} (n - |u_1|)(n - |u_2|)$$

for sufficiently large n , these imply

$$\beta_2 = B, \quad \alpha_3 = \frac{1}{3} + \frac{d(d+1)}{2} - \tau. \quad (12)$$

The coefficient extraction in Section 5 then gives the stronger identity

$$c_1(\theta) = \frac{d+1}{2} \theta + B\theta^2 - \left(\frac{d+1}{2} + \tau - \frac{1}{3} \right) \theta^3. \quad (13)$$

3 Exact connected-support polynomials

For a finite graph H , write $L_j(H) = [z^j] \log I_H(z)$. For a nonempty set of vertices T define

$$w_j(T) = \sum_{U \subseteq T} (-1)^{|T|-|U|} L_j(G_n[U]). \quad (14)$$

The empty graph contributes zero. Then

$$w_j(T) = 0 \quad \text{if } |T| > j \text{ or } G_n[T] \text{ is disconnected,} \quad b_j(n) = \sum_{\emptyset \neq T \subseteq V_n} w_j(T). \quad (15)$$

Indeed, give each vertex its own indeterminate in the independence polynomial. A degree- j monomial of the formal logarithm uses at most j variables. Möbius inversion (14) selects precisely the monomials whose support equals T , after all variables are equated. If the induced graph on T is disconnected, its independence polynomial factors over components, so its logarithm contains no monomial using all of T . This proves (15).

Enumerate nonempty finite connected lattice sets T of size at most j modulo translations. Normalize each representative so that both coordinate minima are zero; do not quotient by rotations or reflections. Let

$$w(T) = \max_{(x,y) \in T} x, \quad h(T) = \max_{(x,y) \in T} y.$$

Its number of translates fitting in V_n is exactly $(n - w(T))_+(n - h(T))_+$. Hence the following identity holds for every n :

$$b_j(n) = \sum_T w_j(T)(n - w(T))_+(n - h(T))_+. \quad (16)$$

There are finitely many supports because the move set is fixed. If

$$R = \max_{u \in S} \max(|u_1|, |u_2|),$$

connectedness implies $w(T), h(T) \leq R(j - 1)$. For $n \geq R(j - 1) + 1$, equation (16) is the exact quadratic (8), with

$$\alpha_j = \sum_T w_j(T), \quad \beta_j = - \sum_T w_j(T)(w(T) + h(T)), \quad \gamma_j = \sum_T w_j(T)w(T)h(T). \quad (17)$$

These are finite rational sums, not numerical fits or polynomial interpolation.

4 Uniform analytic control

A uniform bound is needed to turn the finite-order data into an expansion for a growing number of pieces.

Lemma 1. *For every graph G of maximum degree at most d , its independence polynomial has no zero on*

$$|z| \leq \rho := \frac{1}{4(d+1)}.$$

The logarithm normalized by $\log I_G(0) = 0$ satisfies

$$|\log I_G(z)| \leq 4|V(G)||z|, \quad |[z^j] \log I_G(z)| \leq 4|V(G)|\rho^{1-j}. \quad (18)$$

Proof. Put $\delta = 1/[2(d+1)]$. Induct on the number of vertices, proving that I_G/I_{G-v} exists and differs from 1 in modulus by at most δ . The deletion identity gives

$$\frac{I_G}{I_{G-v}} = 1 + z \frac{I_{G-N[v]}}{I_{G-v}}.$$

Delete the at most d neighbors of v one at a time in $G - v$. The induction hypothesis bounds the second quotient by $(1 - \delta)^{-d} < 2$, since

$$d[-\log(1 - \delta)] \leq \frac{d\delta}{1 - \delta} = \frac{d}{2d+1} < \frac{1}{2}.$$

Thus the ratio differs from 1 by at most $2|z| \leq \delta < 1$, closing the induction and proving nonvanishing. Its analytic logarithm has modulus at most

$$\frac{2|z|}{1 - \delta} \leq 4|z|.$$

Telescope over a vertex deletion order. Cauchy's coefficient estimate on $|z| = \rho$ proves the second bound. $\square$

Fix $R_0 > 1$ and a compact positive interval for θ . The omitted tail after cluster order J in $\log I_n(\theta t/n)$, uniformly on $|t| \leq R_0$, is bounded by

$$\sum_{j>J} 4n^2 \rho^{1-j} (\theta R_0/n)^j = O(n^{1-J}). \quad (19)$$

Use (8) for the finitely many retained indices. Taking $J = M + 2$ and exponentiating gives

$$I_n(\theta t/n) e^{-\theta n t} = e^{a\theta^2 t^2} \left(\sum_{r=0}^M H_r(t; \theta) n^{-r} + O(n^{-M-1}) \right) \quad (20)$$

uniformly on the same disk. Exponentiation preserves the order because the retained expressions are uniformly bounded there. Section 5 identifies H_r and proves that coefficient extraction preserves this error.

5 A finite all-order extraction formula

For $\ell \geq 1$ set

$$h_\ell(t; \theta) = \alpha_{\ell+2}(\theta t)^{\ell+2} + \beta_{\ell+1}(\theta t)^{\ell+1} + \gamma_\ell(\theta t)^\ell, \quad (21)$$

with missing indices below 2 interpreted as zero. Define polynomials H_r by

$$\sum_{r \geq 0} H_r(t; \theta) \epsilon^r = \exp \left(\sum_{\ell \geq 1} h_\ell(t; \theta) \epsilon^\ell \right).$$

Equivalently, $H_0 = 1$ and

$$r H_r = \sum_{\ell=1}^r \ell h_\ell H_{r-\ell}. \quad (22)$$

Define rational polynomials $P_q(m)$ by

$$\sum_{q \geq 0} P_q(m) \epsilon^q = \exp \left(- \sum_{\ell \geq 1} \frac{\epsilon^\ell}{\ell} \sum_{i=0}^{m-1} i^\ell \right). \quad (23)$$

Faulhaber's formula gives $\deg P_q \leq 2q$. The first three are

$$P_0 = 1, \quad P_1 = -\frac{m(m-1)}{2}, \quad P_2 = \frac{m(m-1)(m-2)(3m-1)}{24}.$$

Write $D = t \, d/dt$.

Proposition 1. *The coefficients in Theorem 2 are*

$$c_r(\theta) = e^{-a\theta^2} \sum_{q=0}^r \theta^{-q} P_q(D) \{ e^{a\theta^2 t^2} H_{r-q}(t; \theta) \} \Big|_{t=1}. \quad (24)$$

In particular $c_r = c_r(1) \in \mathbb{Q}$.

Proof. For $k = \theta n$ let $H_{n,k}(t) = I_n(\theta t/n)e^{-kt}$. Expanding e^{kt} and extracting $[t^k]$ gives the exact identity

$$\frac{a(n, k)}{n^{2k}/k!} = \sum_{m=0}^k [t^m] H_{n,k}(t) \frac{(k)_m}{k^m}. \quad (25)$$

The scale in $I_n(\theta t/n)$ is $(k/n^2)^k$, which accounts for the normalization.

For an analytic function f on $|t| \leq R_0$, Cauchy's estimate and $0 \leq (k)_m/k^m \leq 1$ show that the functional on the right of (25) has norm at most $(1 - 1/R_0)^{-1}$ in the disk supremum norm. Thus the analytic remainder in (20) remains $O(n^{-M-1})$ after extraction.

It remains to expand the finite falling product. For $0 \leq m \leq \sqrt{k}$, the elementary symmetric functions in $0, 1, \dots, m-1$ give

$$\frac{(k)_m}{k^m} = \sum_{q=0}^M P_q(m) k^{-q} + E_{M,m,k}.$$

Put $A = m(m-1)/2$. Since the degree- q elementary symmetric function is at most $A^q/q!$,

$$|E_{M,m,k}| \leq \frac{k^{-M-1} A^{M+1}}{(M+1)!} e^{A/k} \leq C_M k^{-M-1} m^{2M+2}. \quad (26)$$

The coefficients of each analytic function in (20) decay geometrically, so summing (26) gives $O(k^{-M-1})$. For $m > \sqrt{k}$ the exact factor is bounded by 1 up to k and vanishes beyond k , while $P_q(m)$ has fixed polynomial growth. These tails, including the extension of each polynomial sum to all m , are $O(\text{poly}(k) R_0^{-\sqrt{k}})$. Finally,

$$\sum_{m \geq 0} m^j [t^m] f(t) = D^j f(1).$$

Apply this identity to the retained terms and use $k^{-q} = \theta^{-q} n^{-q}$. Since k and n are uniformly comparable, (24) and the uniform remainder follow. Exponential factors cancel in (24); the coefficients are rational at $\theta = 1$. $\square$

Theorem 2, and hence the existence assertion of Theorem 1, is now proved. Inserting (12) into the first operator gives (13) and (11).

6 Exact coefficients and reproducibility

For kings and knights the quadratic cluster coefficients through $j = 6$ are as follows. Each row means $b_j(n) = \alpha_j n^2 + \beta_j n + \gamma_j$ once $n \geq R(j-1) + 1$.

j	Kings			Knights		
	α_j	β_j	γ_j	α_j	β_j	γ_j
2	-9/2	6	-2	-9/2	12	-8
3	97/3	-76	44	109/3	-164	180
4	-1133/4	907	-715	-1489/4	2226	-3248
5	13856/5	-10936	10632	21771/5	-31468	55684
6	-58193/2	134242	-458096/3	-110719/2	460910	-2826242/3

The numbers of connected supports modulo translations, by sizes $1, \dots, 6$, are

kings	1	4	20	110	638	3832
knight	1	4	28	234	2162	20972

Inserting the table into (24) proves the explicit coefficients in (2)–(3). Formal division and logarithms give (4)–(6).

The first two uniform-density corrections, beyond (13), are

$$\begin{aligned}
c_1^K(\theta) &= -\frac{\theta}{6}(49\theta^2 - 36\theta - 27), \\
c_2^K(\theta) &= \frac{\theta}{72}(2401\theta^5 - 3528\theta^4 - 2628\theta^3 + 4248\theta^2 + 891\theta - 432), \\
c_1^N(\theta) &= -\frac{\theta}{6}(25\theta^2 - 72\theta - 27), \\
c_2^N(\theta) &= \frac{\theta}{72}(625\theta^5 - 3600\theta^4 + 3924\theta^3 + 7632\theta^2 - 405\theta - 864).
\end{aligned}$$

6.1 Two neighboring board models

For wazirs (orthogonal nearest-neighbor moves, A201511) and ferses (diagonal nearest-neighbor moves, A201861), the carrier is $e^{-5/2}n^{2n}/n!$. Their first four relative coefficients are

	c_1	c_2	c_3	c_4
wazirs	7/3	32/9	-1003/810	68291/9720
ferses	13/3	128/9	8267/810	-388861/9720

These are instances of the same theorem, with the same all-fixed-order error guarantee.

6.2 Finite computation and independent checks

The accompanying `compute_clusters.py` grows every connected support from the singleton and applies (14) and (17). Every finite connected graph has an ordering obtained by deleting leaves of a spanning tree, so every support occurs in this growth procedure. Translation normalization alone avoids erroneous symmetry division. Coefficients of graph logarithms are computed using the exact integer recurrence

$$jL_j = ji_j - \sum_{q=1}^{j-1} qL_q i_{j-q}, \quad I_H(z) = \sum_{j \geq 0} i_j z^j.$$

The file `derive_expansion.py` implements (22)–(24) using rational symbolic algebra. No sequence fit is used.

The direct finite-board program `verify_finite.py` enumerates independent subsets on 3×3 , 4×4 , and 5×5 boards for all four pieces. All twelve comparisons with the clipped-support formula (16) pass through cluster order six. A separate row-mask dynamic program reproduced both OEIS diagonals through $n = 8$ and checked the cluster polynomials at all applicable board sizes through 13 (111 exact comparisons). A separate Touchard-moment extraction verified the four displayed corrections for both main sequences. These finite checks validate the arithmetic implementation; Sections 3–5 supply the proof for all large n .

[write] **Added 2 October 2026, batch 77: the files in the collection.** In the collection the three producer scripts, `replay.sh`, `build.sh` and the independent-verification programs are in `code/` (the latter renamed `independent_verification-check_coefficients.py` and `independent_verification-row_dp.cpp`); the

two JSON files, `finite_verification.txt`, the frozen row-DP output and the independent result file are in `data/`; `PROOF.md` and `SOURCES.md` are at the report root. All are byte-identical to the delivery; the checksum ledger `SHA256SUMS` (verified, 19 of 19 files, at intake) and the delivered PDF are not shipped. The scripts read and write in the current directory and beside themselves under their delivered names, so they do not run in place; the report's `README.md` explains how to rerun them on a copy. At intake the producers, the twelve finite-board checks and the independent Touchard check were rerun on a copy and reproduced every recorded output (up to line endings); `row_dp.cpp` was not compiled, so the 111 row-DP comparisons were checked against the frozen counts only, as in the delivered replay.

7 Growth-index inversion

Let A_n be either main sequence. Stirling's formula gives, at every fixed order,

$$\log A_n = n(\log n + 1) - \frac{1}{2} \log(2\pi n) + a + \frac{d_1}{n} + \frac{d_2}{n^2} + \cdots, \quad (27)$$

where $a = -9/2$ and

$$d_1 = \frac{9}{4} \quad \text{for kings}, \quad d_1 = \frac{49}{4} \quad \text{for knights}.$$

Replacing n by a positive real variable in a fixed truncation defines a smooth asymptotic model; it does not define combinatorial counts on noninteger boards.

For $y \rightarrow \infty$, set

$$L = \log y, \quad x = \frac{L}{W(eL)}, \quad D = \log x + 2, \quad \delta_0 = \frac{\frac{1}{2} \log(2\pi x) - a}{D}, \quad (28)$$

where W is the positive real Lambert function. For any logarithmic model retaining at least d_1/n , the inverse has

$$n = x + \delta_0 + \frac{\delta_0/2 - \delta_0^2/2 - d_1}{xD} + O\left(\frac{1}{x^2 D}\right). \quad (29)$$

Indeed $x(\log x + 1) = L$. The first correction cancels the logarithmic and constant residual. Taylor expansion at $x + \delta_0$ leaves the term

$$\frac{\delta_0^2/2 - \delta_0/2 + d_1}{x}$$

to first order in x^{-1} ; division by the derivative $D + O(1/x)$ gives (29). The omitted residual is $O(x^{-2})$. Recursive Taylor cancellation or Newton iteration gives every further fixed inverse order.

There is an essential qualification for integer thresholds. The leading equivalent proves eventual strict increase, since

$$\frac{A_{n+1}}{A_n} \sim e^2 n.$$

Let $q_M(y)$ solve the smooth logarithmic model through d_M/n^M . For all sufficiently large y , the threshold

$$N(y) = \min\{n \geq n_0 : A_n \geq y\}$$

is bracketed by the ceilings of

$$q_M(y) \pm O\left(\frac{q_M(y)^{-M-1}}{\log q_M(y)}\right). \quad (30)$$

This follows from the uniform pointwise logarithmic error and the positive derivative of the model. The two ceilings agree away from a shrinking window around integers. No finite-order expansion guarantees unconditional rounding when y is arbitrarily close to a sequence value.

[write] **Added 2 October 2026, batch 77: an instance of repository results.** The inversion of this section is an instance of the inversion apparatus of ProveIt’s transseries volumes, and no novelty is claimed for it. The phase is factorial, so the core is not the linear–logarithmic `p0:thm:lambert-core` but `p0:prop:factorial-core` of the transseries-and-inversion volume [5] with $\kappa = 1$, $d = 1$ and right-hand side $L = \log y$: its solution $\exp(v - 1)$, $v = W_0(eL)$, equals $x = L/W(eL)$ of (28) (since $ve^v = eL$), and its slope $\kappa(v + 1) = \log x + 2$ is the D there. The same core is `t2:eq:unbalanced-core` (with $\kappa = \lambda = 1$) of the combinatorial-transseries volume [6]. The correction (29) is the first two terms of the perturbation-operator series `p0:prop:operator-form` with $\varrho(x) = -\frac{1}{2}\log(2\pi x) + a + d_1/x$: the $m = 1$ term gives $\delta_0 - d_1/(xD)$ and the $m = 2$ term gives $(\delta_0 - \delta_0^2)/(2xD)$, as the write step checked by hand; its coefficients involve $\log x$ through D and δ_0 . The integer bracket (30) is the two-ceiling form of the separation condition, part (2) of `p0:thm:staircase`; part (1) (exact rounding) is not invoked and does not apply, because the truncated model is not an interpolation of (A_n) . The generic staircase arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern an arbitrary monotone function and give no formal status to anything here. What is specific to this report is the forward expansion that feeds the inversion, in particular the constants $a = -9/2$ and $d_1 = 9/4, 49/4$.

Scope of the literature check

The OEIS definitions and cited book pages were inspected on 2 October 2026. Searches for these sequence IDs and higher-order chess expansions, and targeted checks of the current relevant ProveIt catalog, did not locate the specialized formulas displayed here. This is a bounded source check, not proof of absence from all literature. The general canonical cluster framework and the classical leading equivalents are explicitly credited. No convergent infinite asymptotic series or beyond-all-orders completion is claimed.

References

- [1] The OEIS Foundation Inc., [A201513](#), [A201540](#), [A201511](#), and [A201861](#). Accessed 2 October 2026.
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- [3] E. Davies, M. Jenssen, and W. Perkins, *A proof of the Upper Matching Conjecture for large graphs*, [arXiv:2004.06695](#), v2, 29 July 2021, especially Section 3.
- [4] E. Pulvirenti and D. Tsagkarogiannis, *Cluster expansion in the canonical ensemble*, *Communications in Mathematical Physics* **316** (2012), 289–306. [arXiv:1105.1022](#).
- [5] [write] ProveIt, *Transseries and inversion*, the transseries-and-inversion volume (`p0:prop:factorial-core`, `p0:prop:operator-form`, `p0:thm:staircase`), `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`.
- [6] [write] ProveIt, *Combinatorial transseries inverses* (`t2:eq:unbalanced-core`), `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`.